Learning Reusable Options by Decomposing Neural Policies

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Abstract

Options provide a natural form of behavioral abstraction in reinforcement learning, but discovering reusable options from trained neural policies remains difficult. A trained policy may contain many useful behaviors that can be reused in downstream problems, yet the number of candidate subpolicies that a neural network can encode grows exponentially with the number of hidden units. We propose a differentiable approach for extracting reusable options from neural policies. Our key observation is that for piecewise-linear neural networks, selecting a neural subprogram is equivalent to assigning each hidden unit one of three labels: inactive, active, or retained in the computation. This yields a mask space over neural subprograms that can be searched with gradient-based optimization. We further show that reusable options can require default input parameters: neuron masks determine what computation is reused, while input masks determine how that computation is called. Experiments in transfer settings with feedforward and recurrent policies show that the resulting options improve sample efficiency on downstream learning.

1 Introduction

In transfer learning settings, an agent benefits from reusing skills learned in previous tasks. Options provide a natural representation for this kind of reuse: instead of choosing only primitive actions, the agent can also choose temporally extended actions that execute useful behaviors over multiple time steps [Sutton et al., 1999]. Portable options can make downstream learning more sample efficient because they allow the agent to reason with behaviors rather than with individual actions [Konidaris and Barto, 2007]. The difficulty is that useful options are not usually given to the agent.

In this paper, we study the problem of discovering reusable options from trained neural policies. A neural policy that solved a previous task may contain useful behaviors for solving a new task, but reusing the entire policy is often too coarse. The policy may depend on features that are specific to the task in which it was trained. At the same time, the behavior we would like to reuse may be only a small part of the computation performed by a network encoding the policy. This creates a decomposition problem: given a trained neural policy, we would like to extract the parts of the policy that encode reusable behaviors and turn them into options for downstream learning.

This problem is easier to state for programmatic representations. If a solution is written as a program, then reuse can be implemented by extracting functions from previous solutions and storing them in a library. Library-learning methods exploit this idea by compressing solutions into reusable abstractions that can reduce the complexity of solving future tasks [Ellis et al., 2023, Cao et al., 2023, Bowers et al., 2023, Rahman et al., 2024]. Neural policies are also programs in the sense that they compute actions through a composition of elementary operations, but their reusable components are not explicitly named functions. They are entangled with the rest of the network and are therefore difficult to extract.

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Prior work showed that ReLU neural policies can be decomposed into neural trees, whose subtrees correspond to subprograms of the original policy [Alikhasi and Lelis, 2024]. These subprograms can be used as options. This provides a direct connection between neural policy decomposition and option discovery. However, the approach requires enumerating candidate subtrees of the neural tree. The number of candidates grows exponentially with the number of hidden units, which makes exhaustive decomposition infeasible beyond very small networks. As a result, the main challenge is how to search the large space of possible options of trained policies without enumerating them.

We address this challenge by turning neural decomposition into a differentiable masking problem. Our key insight is that extracting a subtree from a neural tree can be represented by assigning a label to each hidden ReLU unit of the original network. A hidden unit can be removed from the extracted computation, forced to behave as active, or kept as part of the computation being reused. Thus, searching over neural subprograms can be implemented as learning masks over hidden units. The discrete masks define the extracted option, while continuous mask parameters allow us to search this space with gradient descent. We call our method DIDEC, for "Differentiable Decomposition".

The decomposition of a policy is not enough to define a reusable option. A reusable behavior also needs to specify how it is called in a new context. Consider a behavior that was learned in one location of an environment but should be reused in another location. The computation encoding the behavior may still depend on location features that were present when the behavior was originally learned. In this case, selecting the right hidden units is not sufficient; the extracted computation may also need default input values that allow it to behave as if it were called in the context where the behavior was learned. DIDEC therefore learns input masks together with hidden-unit masks. Neuron masks determine the subprogram to execute; input masks specify how to call such a subprogram. The learned masks determine reusable neural subprograms. Each subprogram defines the policy of an option, while a simple deterministic termination rule determines how long the option is executed.

We evaluate DIDEC in transfer learning settings with feedforward and recurrent policies. The policies are trained on previous tasks and decomposed into options used by an agent learning a downstream task. Our experiments use domains with both complete and partial observability. The networks we decompose induce option spaces far beyond what enumeration can achieve, yet the options extracted by DIDEC improve sample efficiency compared to baselines that reuse previous policies without decomposition. These results support the hypothesis that trained neural policies can contain reusable behavioral abstractions, and that differentiable neural decomposition can discover these abstractions.

2 Problem Formulation

We consider a transfer learning setting in which an agent has solved a sequence of POMDPs and is evaluated on a new POMDP. We write the j-th POMDP as $P_j = (S_j, A, O, p_j, q_j, r_j, S_{0,j})$, where S_j is the state space, A is the action space, O is the observation space, p_j is the transition function, q_j is the observation function, r_j is the reward function, and $S_{0,j}$ is the initial-state distribution. We assume that the POMDPs share the same action space and observation representation, so that policies trained on previous tasks can be evaluated on observations from other tasks.

Before solving the target POMDP P_i, the agent has access to trained neural policies $\Pi_{\text{train}} = \{\pi_1, \ldots, \pi_{i-1}\}$ and observation-action trajectories $\mathcal{T}_{\text{train}} = \{T_1, \ldots, T_{i-1}\}$ generated by rolling out these policies in their corresponding POMDPs. Each trajectory is a sequence $T_j = \{(o_0, a_0), (o_1, a_1), \ldots, (o_{T_j}, a_{T_j})\}$. Our goal is to use Π_{train} and $\mathcal{T}_{\text{train}}$ to construct a set of reusable options Ω that improves the sample efficiency of learning in the downstream task P_i.

An option is a tuple $\omega = (I_\omega, \pi_\omega, T_\omega)$, where $I_\omega \subseteq O$ is the set of observations in which the option can be invoked, π_ω is the option policy, and T_ω is the termination function [Sutton et al., 1999]. We use call-and-return execution: once an option is selected, the agent follows π_ω until the option terminates. In this work, extracted options are intended to be portable across the target observation space, so we set $I_\omega = O$. We also use deterministic fixed-duration termination: an extracted option runs for a fixed number of primitive steps. Thus, DIDEC learns π_ω, and the fixed termination rule turns the resulting masked neural computation into a temporally extended action.

Given trained neural policies Π_{train} and trajectories $\mathcal{T}_{\text{train}}$ from previously solved POMDPs, our task is to discover a set of options Ω such that an agent learning P_i with action space augmented from A to $A \cup \Omega$ requires fewer interactions to achieve high return than an agent learning with A alone.

3 Neural Policy Decomposition

We first describe how we view a trained neural policy as an object that can be decomposed into candidate option policies. This section focuses on feedforward policies with piecewise-linear activations, in which the connection among neural networks, neural trees, and masks is direct.

3.1 Feedforward Neural Policies

We assume that the policies in Π_{train} are encoded in fully connected feedforward neural networks, such as the one shown in Figure 1 (c). Each layer k has n_k neurons $(1, \dots, n_k)$, with $n_1 = |X|$, where X is the observation vector passed as input to the network. The trainable parameters are the weights and biases between consecutive layers. We denote the weights between layers k and $k+1$ by $W^k \in \mathbb{R}^{n_{k+1} \times n_k}$ and the biases by $B^k \in \mathbb{R}^{n_{k+1}}$. The r-th row of W^k, denoted W^k_r, with the r-th entry of B^k, denoted B^k_r, represent the weights and bias of the r-th neuron of the $(k+1)$-th layer.

We denote by $A^k \in \mathbb{R}^{n_k}$ the vector of values produced in the k-th layer. Here, $A^1 = X$, and A^m is the output of a network with m layers. The pre-activation values of layer k are $Z^k = W^{k-1}A^{k-1} + B^{k-1}$, and the layer output is $A^k = g(Z^k)$, where $g(\cdot)$ is the activation function. For intermediate layers, we consider piecewise-linear activation functions, such as ReLU, where $g(z) = \max(0, z)$. The output layer produces the action logits or action probabilities defining the policy.

Example 1. *The network in Figure 1(c) has $n_1 = 5$ input units, $n_2 = 2$ hidden units, and $n_3 = 1$ output unit. The omitted (zero-weight) connections from x_2 and x_4 are kept as zero columns so that the matrices have the dimensions stated above. The weight matrices and bias vectors are*

$$W^1 = \begin{pmatrix} -1 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix}, \quad B^1 = \begin{pmatrix} 0 \\ 0.5 \end{pmatrix}, \quad W^2 = (-1 \quad -8), \quad B^2 = (5).$$

3.2 Neural Trees and Neural subprograms

A feedforward network with ReLU activations can be mapped to an equivalent neural tree [Alikhasi and Lelis, 2024]. We explain this mapping with the example in Figure 1. The example shows a small ComboGrid environment, a trajectory of observation-action pairs, a neural policy that fits this trajectory, the neural tree equivalent to this policy, and a compressed policy obtained from one masked computation. The default input value shown in the figure previews the input-mask mechanism introduced in Section 4; here, we focus on the neural-tree and hidden-mask correspondence.

Each hidden ReLU neuron is mapped to a level in the neural tree. In Figure 1, neuron A^2_1 is represented by the root level of the tree, A^2_2 by the second level of the tree, and A^3_1 by the tree's leaves. Each internal node considers the two possible outcomes of a ReLU neuron: if $Z \leq 0$, the output is 0 and the computation follows the left branch; if $Z > 0$, the output is Z and the computation follows the right branch. A path from the root to a leaf corresponds to an activation pattern of the network [Montúfar et al., 2014]: the active/inactive state of each neuron for a given input.

Example 2. *We trace two root-to-leaf paths in Figure 1(d) using the weights of Example 1. Each internal node branches on the sign of the pre-activation of one hidden unit: the left branch fixes its output to 0 (inactive), and the right branch fixes its output to its pre-activation value (active).*

Left–right path. *Take the left branch at the root (A^2_1 inactive, so $A^2_1 = 0$) and the right branch at the second level (A^2_2 active, so $A^2_2 = Z^2_2 = -x_5 + 0.5$). The output is*

$$A^3_1 = \sigma\big(-A^2_1 - 8A^2_2 + 5\big) = \sigma(-0 - 8(-x_5 + 0.5) + 5) = \sigma(8x_5 + 1),$$

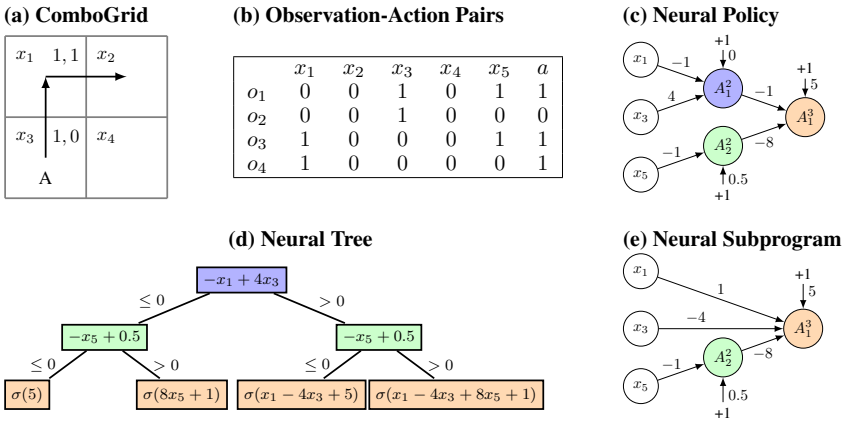
which is the second leaf of Figure 1(d).

Right–left path. *Take the right branch at the root (A^2_1 active, so $A^2_1 = Z^2_1 = -x_1 + 4x_3$) and the left branch at the second level (A^2_2 inactive, so $A^2_2 = 0$). The output is*

$$A^3_1 = \sigma(-(-x_1 + 4x_3) - 8 \cdot 0 + 5) = \sigma(x_1 - 4x_3 + 5),$$

which is the third leaf of Figure 1(d). The remaining two leaves are obtained analogously.

Each subtree of the neural tree provides a subprogram of the policy. In the options view, such a subprogram is a candidate option policy: it maps observations to action probabilities, but it does not by itself specify how long the behavior should be executed. Section 4 describes how DIDEC learns such subprograms from data and wraps them with deterministic termination to produce options.



	x_1	x_2	x_3	x_4	x_5	a
o_1	0	0	1	0	1	1
o_2	0	0	1	0	0	0
o_3	1	0	0	0	1	1
o_4	1	0	0	0	0	1

Figure 1: **(a)** Instance of a ComboGrid environment with combos of length two. The sequence $1, 0$ moves up, while $1, 1$ moves right. Agent 'A' starts at x_3 and reaches x_2 after $1, 0, 1, 1$. **(b)** Observation-action pairs for the trajectory in (a). The observation includes the agent location and bit x_5, set to 1 if no action was taken in a sequence, and 0 otherwise. **(c)** Neural network that fits the data in (b), with ReLU hidden units and a sigmoid output; zero-weight connections are omitted. **(d)** Neural tree equivalent to (c): the left subtree encodes the "right combo" and the right subtree with default $x_3 = 1$ encodes the "up combo". **(e)** Neural subprogram obtained by setting A_1^2 active.

3.3 Subtrees as Hidden-Unit Masks

The neural tree depends on the order in which we branch on hidden units, but its subtrees do not: what a subtree computes depends only on which units have been fixed to 0 (left ancestors), which have been fixed to their pre-activation value (right ancestors), and which remain inside the subtree. This observation lets us replace subtrees with a simpler object. Let $H = \{h_1, \ldots, h_d\}$ be the hidden ReLU units of a feedforward policy π_θ. A hidden-unit mask is a function assigning a label to a unit

$$m : H \to \{\text{inactive}, \text{active}, \text{retained}\},$$

where inactive replaces h's output with 0, active replaces it with its pre-activation value, and retained keeps the original ReLU computation. We write $\mathcal{M}_H(\pi_\theta)$ for the set of all such masks. Since each neuron can be assigned one out of three labels, then $|\mathcal{M}_H(\pi_\theta)| = 3^{|H|}$.

Every subtree induces a mask: ancestors on the left branch are inactive, ancestors on the right branch are active, and units inside the subtree are retained. Conversely, any mask is realized by ordering the inactive and active units first (taking the corresponding branches) and then the retained units, and selecting the subtree rooted where the active/inactive choices end. Subtrees that induce the same mask compute the same function, so the neural-subprogram space is in bijection with $\mathcal{M}_H(\pi_\theta)$.

This correspondence justifies our mask-based approach: searching over masks is equivalent to searching over subtrees of all neural trees of π_θ, and learning continuous mask parameters gives a differentiable relaxation of this discrete search.

3.4 Masked Forward Pass

We represent the masks of the neurons in layer k by a matrix $\Theta^k \in \mathbb{R}^{n_k \times 3}$. The parameters in the i-th row of Θ^k define a distribution over three possible states for neuron i: inactive, active, or retained in the extracted computation. We compute the following, where $M^k_{i,:} \in \{0, 1\}^3$.

$$Z^k = W^{k-1}A^{k-1} + B^{k-1} \quad \text{and} \quad M^k_{i,:} = \text{OneHot}\big(\text{Softmax}(\Theta^k_{i,:})\big)\,,$$

The masked layer output is then

$$A^k = M^k_{:,1} \odot \mathbf{0} + M^k_{:,2} \odot Z^k + M^k_{:,3} \odot \text{ReLU}(Z^k).$$

Rows with a 1 in the first column set the corresponding neuron to 0 and make it inactive; rows with a 1 in the second column replace the ReLU output by the pre-activation value Z^k and make the neuron active; rows with a 1 in the third column keep the original ReLU and retain the neuron's computation.

The function OneHot is non-differentiable because of its implicit maximum. When updating Θ^k with gradient descent, we use the straight-through estimator [Bengio et al., 2013], which passes the gradient computed up to the maximum operation in the backward pass directly to the Softmax layer. We also considered the modified $\tanh$ function proposed by Pitis [2017] and Koul et al. [2019] for discretizing into three values, but preliminary experiments favored the Softmax approach.

Example 3. *The subprogram shown in Figure 1(e) is obtained from the network in Figure 1 (c) by setting A_1^2 active and retaining A_2^2. The subprogram is obtained with the following neuron mask*

$$M^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

where row 1 corresponds to A_1^2 (active, column 2) and row 2 to A_2^2 (retained, column 3). For an observation X, the pre-activation vector is $Z^2 = W^1 X + B^1$, and the masked layer output is

$$A^2 = M^2_{:,1} \odot \mathbf{0} + M^2_{:,2} \odot Z^2 + M^2_{:,3} \odot \mathrm{ReLU}(Z^2) = \begin{pmatrix} Z_1^2 \\ \mathrm{ReLU}(Z_2^2) \end{pmatrix} = \begin{pmatrix} -x_1 + 4x_3 \\ \mathrm{ReLU}(-x_5 + 0.5) \end{pmatrix}.$$

Composing with the output layer gives $A_1^3 = \sigma(-A_1^2 - 8A_2^2 + 5) = \sigma(x_1 - 4x_3 - 8A_2^2 + 5)$, which is exactly the subprogram shown in Figure 1(e). Because A_1^2 is active, its ReLU is no longer needed and the unit can be folded into the output neuron, leaving a network with a single hidden unit.

4 DIDEC: Learning Options from Neural Policy Decompositions

Section 3 defines the mask space induced by the neural-tree decomposition of a feedforward ReLU policy. We now describe how DIDEC learns masks from trajectories generated by previously trained policies and turns the resulting masked subprograms into options. Throughout this section, the weights of the trained policies are fixed. The only learned parameters are the masks.

4.1 From Masked Subprograms to Options

A hidden-unit mask selects a subprogram of a trained policy. Such a subprogram maps observations to action probabilities, so it is the policy of an option. To obtain a temporally extended action, we wrap the masked policy in a repeat loop. If π_θ^Θ denotes the policy obtained by applying mask parameters Θ to a source policy π_θ, then a length-b option has the form $\mathrm{repeat}(b) : \pi_\theta^\Theta$. When this option is selected, the agent executes π_θ^Θ for b primitive steps and then returns control to the high-level policy.

This construction follows the options view of neural decomposition: the extracted subprogram is not transferred as an entire solution to a previous task, but as a reusable behavior that can be invoked by a downstream learner. The next sections describe the parameters that define such a behavior.

4.2 Input Masks as Default Parameters

A reusable option is not specified only by its computation. It also needs to specify how this computation is called in a new context. A subprogram extracted from a policy trained in one environment might depend on input features that describe the context in which the behavior was learned. If those features are read directly from a new environment, the same computation may no longer produce the behavior. In this case, the extracted option needs default input parameters.

DIDEC learns these default parameters by masking the input observation. For binary inputs, we use the same Softmax and one-hot discretization used for hidden-unit masks. Each input coordinate is assigned one of three labels: read the value from the environment, set it to 0, or set it to 1. Let $X \in \{0,1\}^{n_1}$ be the observation and $\Theta^X \in \mathbb{R}^{n_1 \times 3}$ be the input-mask parameters. For each input i,

$$M^X_{i,:} = \mathrm{OneHot}(\mathrm{Softmax}(\Theta^X_{i,:})) \in \{0,1\}^3,$$

with one entry for each possible label. The masked input is then $\widetilde{X}_i = M^X_{i,1} \cdot X_i + M^X_{i,2} \cdot 0 + M^X_{i,3} \cdot 1$.

Example 4. *In this example we combine the neuron mask of Example 3 with an input mask that defaults x_3 to 1 and reads the remaining inputs from the environment. The input mask has*

$$M^X_{3,:} = (0,\, 0,\, 1) \quad \textit{(default to 1)}, \qquad M^X_{i,:} = (1,\, 0,\, 0) \quad \textit{for } i \in \{1,2,4,5\} \textit{ (read from env.)},$$

so that $\widetilde{X} = (x_1,\, x_2,\, 1,\, x_4,\, x_5)$*. The masked policy from Example 3, evaluated on* $\widetilde{X}$*, computes*

$$\pi_\theta^\Theta(\widetilde{X}) \;=\; \sigma\big(x_1 - 4 - 8\,\mathrm{ReLU}(-x_5 + 0.5) + 5\big).$$

Suppose the agent is at location x_1 *in a new environment, so the actual observation has* $x_1 = 1$ *and* $x_3 = 0$*, and consider the second step of the up combo, where* $x_5 = 0$*. Then* $\mathrm{ReLU}(-x_5 + 0.5) = 0.5$*. Without the input mask, the masked policy (using* $x_3 = 0$ *instead of the default* 1*) evaluates to* $\sigma(1 - 0 - 4 + 5) = \sigma(2) \approx 0.88$*, predicting action* 1 *rather than the action* 0 *required by the up combo. With the input mask,* $\sigma(1 - 4 - 4 + 5) = \sigma(-2) \approx 0.12$*, correctly predicting action* 0*. Setting* $x_3 = 1$ *lets the agent "pretend" it is in* x_3*, so the up combo generalizes to other locations.*

For continuous inputs, the same idea can be implemented by learning whether each coordinate should be read from the environment or replaced by a learned default value. In this case, each input mask has two labels. Let $\Theta^X \in \mathbb{R}^{n_1 \times 2}$ be the mask parameters and let $\Theta^D \in \mathbb{R}^{n_1}$ be the parameters of the default input values. We compute the following, where g is chosen to match the range of the input coordinate. For example, if the input values are in $[-1, 1]$, one can use $g = \tanh$.

$$M^X_{i,:} = \mathrm{OneHot}(\mathrm{Softmax}(\Theta^X_{i,:})) \in \{0,1\}^2 \quad \text{and} \quad \widetilde{X}_i = M^X_{i,1} \cdot X_i + M^X_{i,2} \cdot g(\Theta^D_i),$$

Although input masking can sometimes be sufficient to obtain a reusable option, hidden-unit masking has an additional benefit: it compresses the extracted policy. A hidden unit masked as inactive can be removed by substituting its output with zero, and a hidden unit masked as active can be removed by substituting its output with its pre-activation value. The compressed policy in Figure 1 (e) shows this effect. Once neuron A^2_1 is masked as active, its ReLU is no longer needed, and the output neuron can be rewritten directly as a function of the inputs and A^2_2, namely $A^3_1 = \sigma(x_1 - 4x_3 - 8A^2_2 + 5)$. Thus, A^2_1 can be removed from the extracted model. More generally, every neuron at layer k that is masked as active or inactive can be eliminated by rewriting the affine functions of neurons in layer $k + 1$. In this sense, neuron masking does not only select reusable option policies; it also produces smaller policies by removing computation that is fixed by the extracted subprogram.

For recurrent policies, an extracted option must also specify the recurrent state from which execution starts. DIDEC learns the initial hidden state, and for LSTMs also the initial memory state, together with the input masks. This can be viewed as selecting the initial mode of the recurrent policy before the option is executed. Appendix B describes the full parameterization in detail.

4.3 Learning Masked Option Policies with Behavior Cloning

We use the trajectories generated by the policies in Π_{train} to learn masks by behavior cloning [Schaal, 1997]. Let $\mathcal{T}_{\mathrm{train}} = \{T_1, \ldots, T_{i-1}\}$ be the trajectories obtained by rolling out the trained policies in their corresponding POMDPs. For each trajectory T_j, DIDEC considers sub-trajectories of observation-action pairs with length $z \in \{2, 3, \ldots, z_{\max}\}$, where $z_{\max}$ is a hyperparameter.

Each sub-trajectory is used as a short behavioral specification. Given a sub-trajectory

$$\tau = ((o_t, a_t), (o_{t+1}, a_{t+1}), \ldots, (o_{t+z-1}, a_{t+z-1}))$$

from trajectory T_j, we learn masks for a source policy $\pi_\ell \in \Pi_{\mathrm{train}}$ with $\ell \neq j$. The condition $\ell \neq j$ prevents the method from simply recovering the policy that generated the sub-trajectory. Instead, it asks whether a different trained policy contains a subprogram that can reproduce the behavior in τ, which encourages the extraction of behaviors that are reusable across tasks.

For a fixed source policy π_ℓ and sub-trajectory τ, DIDEC minimizes the cross-entropy loss $-\sum_{(o,a)\in\tau} \log \pi^\Theta_\ell(a \mid o)$, where π^Θ_ℓ denotes the source policy with the current input masks, hidden-unit masks, and recurrent-state parameters, when applicable. The weights of π_ℓ are frozen; gradient descent updates only Θ. After training, the masked policy π^Θ_ℓ is wrapped as an option of length $|\tau|$. Longer sub-trajectories generate longer options, but they also tend to be more specialized to the source task. The hyperparameter $z_{\max}$ therefore controls the trade-off between the number and duration of generated options and the risk of extracting overly specific behaviors.

4.4 Candidate Option Extraction

Algorithm 1 summarizes the option-extraction procedure. The algorithm returns a candidate option set Ω_{cand}. In our experiments, this candidate set is pruned before downstream learning so that the

Algorithm 1 DIDEC's Neural Decomposition with Behavior Cloning

Require: Policies Π_{train}, trajectories $\mathcal{T}_{\text{train}}$, maximum length $z_{\max}$, epochs E, learning rate α
Ensure: Candidate option set Ω_{cand}

```
Ω_cand ← ∅
for each sub-trajectory τ of length 2, ..., z_max in T_train do
    Let j be the index of the policy that generated τ
    for each source policy π_ℓ ∈ Π_train with ℓ ≠ j do
        Initialize mask/default parameters Θ
        Train Θ for E steps to minimize −Σ_{(o,a)∈τ} log π_ℓ^Θ(a | o)
        Ω_cand ← Ω_cand ∪ {repeat(|τ|) : π_ℓ^Θ}
return Ω_cand
```

high-level agent does not need to choose among too many options. We use the subset selection process of Alikhasi and Lelis [2024], which we describe in Appendix J for the interested reader.

Let $T_{\max}$ be the length of the longest trajectory in $\mathcal{T}_{\text{train}}$. For each trajectory, DIDEC considers $O(T_{\max} z_{\max})$ sub-trajectories. Each sub-trajectory is paired with each source policy that did not generate it. Thus, since $N = |\mathcal{T}_{\text{train}}| = |\Pi_{\text{train}}|$, the number of mask-learning problems is $O(T_{\max} z_{\max} N^2)$. If each mask-learning problem uses E gradient steps and the source policy has W parameters, the cost of generating Ω_{cand} is $O(T_{\max} z_{\max} N^2 E W)$. The important point is that this cost does not include the exponential number of neural-tree subprograms. DIDEC searches this space by optimizing mask parameters rather than enumerating candidate subtrees. We also provide a standard finite-class generalization bound for the mask space, which is given in Appendix K.

5 Related Work

Options Early work relied on manually designed options [Sutton et al., 1999], but many methods now learn options automatically—though they often require human choices such as the number of options [Bacon et al., 2017, Igl et al., 2020] or their duration [Frans et al., 2017, Tessler et al., 2017]. DIDEC avoids such supervision but depends on data from previously solved tasks. While options have also been explored for improving exploration [Jinnai et al., 2020, Machado et al., 2023] and faster credit assignment [Mann and Mannor, 2014], DIDEC focuses on extracting functions that encode reusable behaviors to facilitate downstream knowledge transfer [Konidaris and Barto, 2007].

Transfer Learning Knowledge transfer across tasks has been studied via regularization [Kirkpatrick et al., 2017], architectural priors [Rusu et al., 2016, Yoon et al., 2017, Schwarz et al., 2018], and experience replay [Rolnick et al., 2019]. A common strategy is to reuse parts of pretrained models [Clegg et al., 2017, Shao et al., 2018]. Unlike these approaches, DIDEC transfers knowledge by extracting reusable sub-programs (options) through gradient-based network decomposition.

Library Learning DIDEC is a library-learning method [Cao et al., 2023, Bowers et al., 2023, Rahman et al., 2024, Palmarini et al., 2024]. As a representative example, DreamCoder [Ellis et al., 2023] builds a library of reusable lambda calculus functions to solve supervised learning problems. Similarly, DIDEC constructs a library of reusable programs, but the programs are extracted from neural networks and the underlying problem is reinforcement learning, and not supervised learning. DIDEC contributes toward bridging symbolic and neural paradigms in library-learning methods.

Masking Networks Masking has been used in transfer learning, e.g., SupSup [Wortsman et al., 2020] and Modulating Masks [Ben-Iwhiwhu et al., 2022], but these approaches mask weights, not neurons or input features. In contrast, DIDEC masks neurons with piecewise-linear activations, allowing for the extraction of subprograms. Input masking has also been explored—for mitigating visual distractions [Bertoin et al., 2022, Grooten et al., 2024] or learning auxiliary tasks [Yu et al., 2022]—but DIDEC uses input masking to define default parameters to enable generalization.

6 Empirical Evaluation

We hypothesize that DIDEC can extract options that generalize to downstream tasks. To evaluate our hypothesis, we use Advantage Actor-Critic (A2C) [Mnih et al., 2016] and Proximal Policy

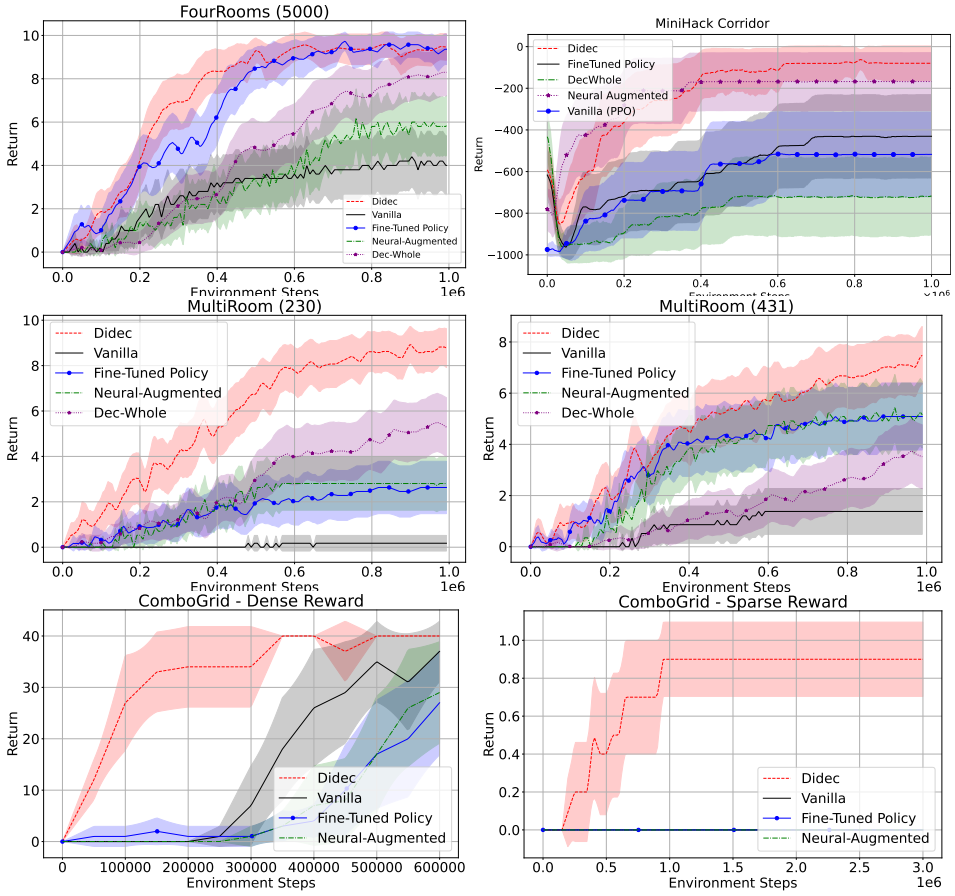


Figure 2: Learning curves of DIDEC and baselines on downstream tasks with temporally extended options. Curves show average return with 95% confidence intervals, using 50 seeds for FourRooms, 57 for MultiRoom, 10 for ComboGrid, and 22 for MiniHack.

Optimization (PPO) [Schulman et al., 2017] with either a feedforward network encoding the policy or an LSTM, depending on whether the problem is partially observable or not. Our network sizes range from 64 to 256 neurons, resulting in a range of 3.43×10^{30} to 1.39×10^{122} sub-trees, which far exceeds the capacity of enumerative approaches. The architectures we use are detailed in Appendix E.

Our experiments are on MiniGrid [Chevalier-Boisvert et al., 2023], MiniHack's Corridor [Samvelyan et al., 2021], ComboGrid [Alikhasi and Lelis, 2024], and Meta-World's "Push the Puck" [Yu et al., 2021]. We use A2C in our MiniGrid experiments and PPO in MiniHack, ComboGrid, and Meta-World. Our choice of learning algorithm is arbitrary, and it illustrates the applicability of DIDEC. Neuron masking can be used when the activation functions are piecewise linear, and input masking can be used when the input domains are finite, which includes continuous inputs, as in Meta-World. We use feedforward networks with MiniGrid, MiniHack, and Meta-World, and an LSTM with ComboGrid. A complete description of the environments used is given in Appendix L.

Alikhasi and Lelis [2024] compared an enumerative version of DIDEC to a range of option-learning methods, including Modulating Masks [Ben-Iwhiwhu et al., 2022], Progressive Neural Networks [Rusu et al., 2016], Option-Critic [Bacon et al., 2017], ez-greedy [Dabney et al., 2021], and DCEO [Klissarov and Machado, 2023], and showed a clear advantage for the decomposition approach. That advantage, shared by DIDEC, arises from leveraging policies trained on $\{P_j\}_{j=1}^{i-1}$ to improve sample efficiency on P_i, which previous methods were not designed to leverage.

To evaluate DIDEC, we compare it against baselines that also exploit previously learned policies. The first augments the action space with one-step options derived from earlier policies (**Neural-Augmented**). The second is like DIDEC except it reuses prior policies as a single undivided function

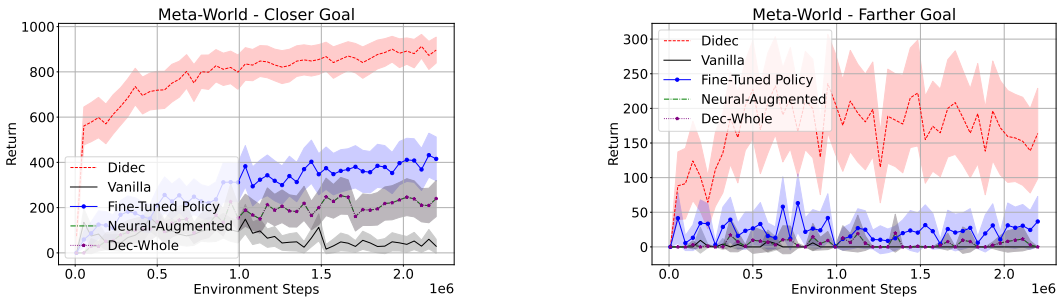


Figure 3: Meta-World results: average return with 95% confidence intervals over 97 seeds.

rather than decomposing them (**Dec-Whole**). The third is also like DIDEC except that it fine-tunes the policy weights instead of training masks (**Fine-Tuned Policy**). We also use the base learning algorithm used with options (either A2C or PPO), but without options (**Vanilla**).

The results with these baselines also serve as an ablation study for DIDEC. Neural-Augmented does not consider neural decomposition and temporal abstraction; Dec-Whole does not consider neural decomposition; Fine-Tuned Policy trains the neural network weights rather than the mask weights. Together, these baselines evaluate different components of DIDEC. We present other ablations of DIDEC in Appendices D and G. Appendix C shows that some of the options DIDEC learns are interpretable, while Appendix H shows how DIDEC's learned options can help with exploration in downstream tasks. Finally, Appendix I presents a running time analysis of DIDEC.

For each option-based approach, we augment the agent's action space with the set of options that each method selects. Then, we train a policy on the test problem of each domain, an action-augmented agent with either A2C or PPO. In Appendix F, we present the values considered in our hyperparameter sweep for DIDEC and baselines. We considered masking neurons, input features, or both, as hyperparameters of DIDEC. Our empirical results are presented in plots, where the y-axis represents the average return and the x-axis represents the number of agent interactions.

Figure 2 and 3 present the results. The results support our hypothesis: DIDEC is competitive across all domains and substantially more sample-efficient in several of them. The performance of DIDEC on MultiRoom (230) is more evident than on MultiRoom (431) and MiniHack. We conjecture that this occurs because one of the rooms in problem 230 is much narrower than those in the training problems (see Appendix L), making it more difficult to generalize. The structures encountered in the training of MultiRoom and MiniHack are more similar to those found in the test problem, thus allowing the baselines also to discover options that generalize. DIDEC also presents similar advantages over the tested baselines on the continuous Meta-World tasks, where baselines fail to learn any meaningful policies. The results on ComboGrid (dense and sparse) demonstrate that DIDEC is also applicable to recurrent models. In this domain, Dec-Whole did not select any options, so it behaves like Vanilla.

7 Conclusions and Limitations

We introduced DIDEC, a differentiable method for discovering reusable options from trained neural policies. DIDEC replaces exhaustive enumeration of neural-tree subprograms with mask optimization: hidden-unit masks select reusable computation, and input-default masks specify how that computation is invoked in new contexts. This formulation connects neural policy decomposition to option discovery. Experiments with feedforward and recurrent policies show improved downstream sample efficiency, while Meta-World suggests that the same masking idea can also be useful in continuous-control settings. These results support the view that trained neural policies can contain reusable behavioral abstractions that differentiable decomposition can expose to downstream learners.

While we make important progress in extracting reusable subprograms from entangled neural representations, a few limitations remain. DIDEC assumes access to trained policies with compatible observation and action spaces. Also, the continuous-control variant does not yet instantiate temporally extended options. While we showed how the decomposition can work with both binary and continuous features, more research is needed to handle richer inputs, such as pixels. Future work will study online decomposition schemes to discover options while learning within a task.

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A Societal Impact

This paper advances basic research in sequential decision making. Our experiments are conducted in controlled benchmark domains, and we do not expect the specific systems evaluated here to have immediate societal impact. In the long run, however, methods for discovering reusable behavioral abstractions could have dual-use implications, since they may be applied to sequential decision-making systems more broadly. Potential benefits include improving sample efficiency and transfer in autonomous agents, while potential risks include enabling more capable autonomous behavior in settings where deployment is unsafe or insufficiently supervised.

B Recurrent option policies

In partially observable problems, policies are often represented by recurrent models such as LSTMs [Hochreiter and Schmidhuber, 1997] and GRUs [Chung et al., 2014, Bahdanau et al., 2015]. When extracting options from recurrent policies, input masks alone are insufficient. We must also specify the recurrent state from which the option starts. Recurrent policies can be viewed as approximating finite-state machines, where the hidden state represents the current mode of the underlying machine. An option extracted from a recurrent policy therefore needs an initial mode.

DIDEC					Fine-Tuned Policy				
0.6	0.6		0.6	0.6	0.0	0.0		0.0	0.0
0.6	0.6	0.6	0.6	0.6	0.0	0.0	0.2	0.0	0.0
	0.6	0.6	0.6			0.0	0.0	0.0	
0.6	0.6	0.6	1.0	0.6	0.0	0.0	0.0	0.0	0.0
0.6	0.6		0.6	0.6	0.0	0.0		0.0	0.0

Table 1: Generalization results of options generated by DIDEC and Fine-Tuned Policy in ComboGrid.

DIDEC learns this initial mode together with the input masks. For the initial hidden vector $h_0 \in [-1, 1]^r$, DIDEC learns parameters Θ^h and sets

$$h_0 = \tanh(\Theta^h),$$

where Θ^h has the same shape as h_0. For LSTMs, we also learn the initial memory vector c_0. Since c_0 is unbounded, we parameterize it with two tensors Θ^s and Θ^m, representing sign and magnitude

$$c_0 = \tanh(\Theta^s) \odot \mathrm{SoftPlus}(\Theta^m).$$

When the option is invoked, the recurrent policy is initialized with the learned hidden and memory vectors and then executed for the duration of the option. In this way, learning the recurrent state selects the option's initial mode, while input masks control which observations are allowed to define the behavior from that mode.

C Examples of Options Learned in ComboGrid

Figure 4 shows the options learned in ComboGrid. In this domain, the learned options are modular and encode well-defined behaviors in the underlying LSTM, such as moving forward twice and turning right. The agent learned options that are independent of the observation, which is helpful in this domain, since the action combination does not change based on the observation. The agent can solve the test problem using only options.

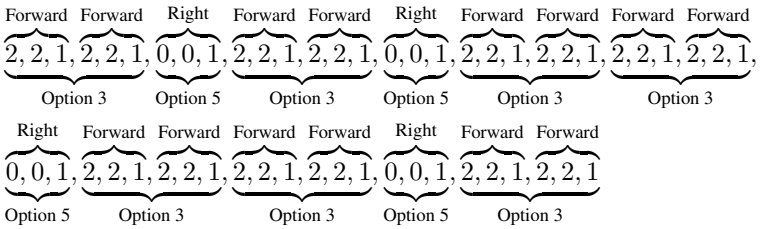


Figure 4: Sequence of actions DIDEC's agent learned on ComboGrid. Labels at the top describe the semantics of a combo, while those at the bottom describe the options learned.

Each cell in Table 1 shows the fraction of DIDEC's (top) and Fine-Tuned Policy's (bottom) options that generalize to the unseen observations of the test problem; the gray cells denote goal locations, so we did not calculate their values. At least 60% of DIDEC's options, which encode helpful combos such as 'move, move', generalize, while Fine-Tune's options fail. Recall that Fine-Tune differs from DIDEC only in the weights it adapts—Fine-Tune adjusts model weights, whereas DIDEC learns masks. This result illustrates how extracting subprograms with default parameters can enable generalization.

D Varying the Training Environments

Random training instances. The training problems used in our main experiments were chosen arbitrarily, which raises the question of whether DIDEC's advantage depends on a specific choice. To check this, we re-ran MultiRoom with three randomly selected training problems and evaluated on the test problems with seeds 230 and 431. Figure 5 shows the result: DIDEC's sample efficiency advantage over the baselines is similar to what we report in Figure 2.

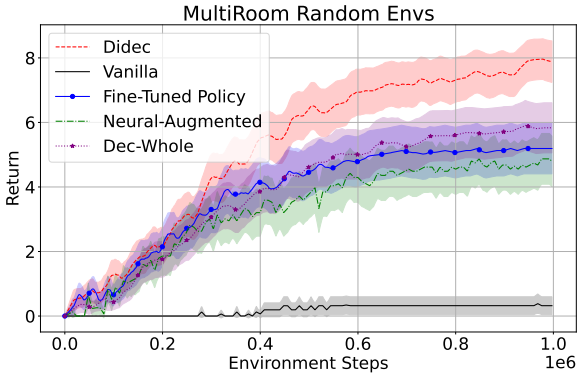


Figure 5: Learning curves on MultiRoom test problems 230 and 431 when the three training problems are selected randomly.

Size of the training set. The quality of the options DIDEC learns depends on the problems $\{P_j\}_{j=1}^{i-1}$ used for training. Because the masking objective extracts a subprogram of one source policy that reproduces a sub-trajectory generated by a different source policy, a behavior must appear in at least two training problems for DIDEC to extract a generalizing option for it. Adding a problem whose behaviors do not appear elsewhere enlarges the candidate pool without adding new reusable options.

ComboGrid makes this concrete. Figure 6 compares DIDEC with two, three, and four training problems. With two problems, the trajectories share the move-forward combo $(2, 2, 1)$ and the turn-right combo $(0, 0, 1)$, and DIDEC extracts options for both. The third problem adds the turn-left combo $(1, 1, 0)$, but it appears in only one trajectory, so no generalizing option is extracted for it; the candidate pool grows from $|\Omega| \approx 150$ to ≈ 450 without adding reusable behaviors. The fourth problem also contains $(1, 1, 0)$, completing coverage of all three combos and yielding the best average sample efficiency.

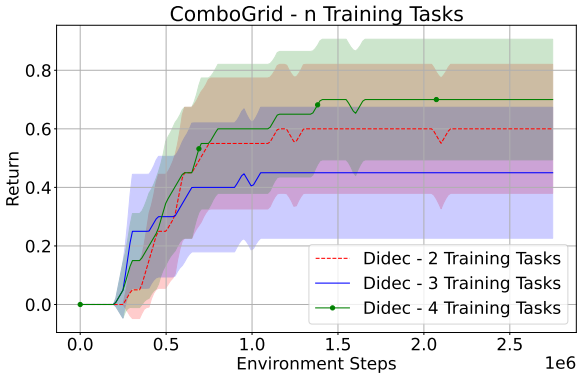


Figure 6: DIDEC's performance on ComboGrid (sparse reward) with two, three, and four training problems. Curves show average return with 95% confidence intervals over 20 seeds.

The drop at three tasks also reflects a selection effect: the larger candidate pool makes the local-search subset selection less reliable under a fixed budget of restarts. We observed the same pattern in MultiRoom, when increasing the training set from three to six problems, where the smaller set already covered the behaviors needed at test time. Increasing the search budget can mitigate this, and improving the scalability of the selection procedure is a natural direction for future work.

E Neural Architectures

We adopt a neural Actor-Critic framework. The model is composed of two networks with shared input: an actor that parameterized a stochastic policy, and a critic that estimates the value function.

Let d_{obs} denote the dimensionality of the observation vector (after flattening), and let $|\mathcal{A}|$ be the number of discrete actions.

E.1 Feedforward Architecture

The following network was used in FourRooms and MultiRoom. The architecture used with MiniHack is identical, except that it uses 256 neurons instead of 64.

Actor Network. The actor network maps an input observation to a distribution over actions via the following architecture:

- A linear layer with input size d_{obs} and output size 64,
- ReLU activation,
- A final linear layer with output size $|\mathcal{A}|$ producing action logits.

The final layer is initialized with a reduced standard deviation (std $= 0.01$) to promote stability during early exploration.

Critic Network. The critic network has a deeper architecture to estimate the state value:

- A linear layer from d_{obs} to 64 units,
- ReLU activation,
- Another linear layer with 64 hidden units,
- ReLU activation,
- A final linear layer mapping to a scalar value.

Weight Initialization. Weights are initialized orthogonally with a gain factor of $\sqrt{2}$, and biases are set to 0.0. The final layer of the critic is initialized with standard deviation of 1.0.

E.2 LSTM Architecture

We use the Stable-Baselines3 (SB3) [Raffin et al., 2021] implementation of PPO with an LSTM representation. The policy consists of: (i) two unidirectional LSTMs, one for the actor and another for the critic, with 16 neurons that produce a latent state; and (ii) separate, feedforward network heads for actor and critic.

Actor Network. From the last LSTM output at each timestep, the actor head applies:

- A linear layer with 32 units,
- $\tanh$ activation,
- A final linear layer with output size $|\mathcal{A}|$ producing action logits.

Critic Network. The critic shares the same LSTM core but uses its own head:

- A linear layer with 32 units,
- $\tanh$ activation,
- A final linear layer mapping to a scalar value.

Weight Initialization. Weights are initialized orthogonally with a gain factor of $\sqrt{2}$, and biases are set to 0.0. The final actor layer is initialized with a smaller weight scale of 0.01. The final critic layer is initialized with a gain of 1.0.

Hyperparameter	Values Tested
Learning rate	{0.01, 0.005, 0.001, 0.0005, 0.00005}
Clipping coefficient	{0.05, 0.1, 0.15, 0.2, 0.3}
Entropy coefficient	{0.01, 0.02, 0.03, 0.05, 0.1, 0.2}
Training time steps	{1 million}

Table 2: Hyperparameters evaluated for DIDEC on FourRooms (5000).

Methods	Learning rate	Clipping coefficient	Entropy coefficient
Didec	0.005	0.3	0.02
Fine-Tune	0.005	0.3	0.03
Dec-Whole	0.001	0.2	0.03
Neural-Augmented	0.001	0.3	0.03
Vanilla	0.0005	0.3	0.03

Table 3: Chosen hyperparameters per method on FourRooms (5000).

F Hyperparameter Sweeps

The hyperparameter sweeps were performed with a grid search over the values shown in the tables of this section. We considered three seeds for each set of values, and the value of a set is the average area under the curve (AUC) of the agent's return across the three seeds. We report the result of each approach using the hyperparameter set that has the largest average AUC. We consider the $z_{\max}$ of 5 and 10 across different benchmarks. Experiments were performed on Intel Xeon Gold 6448Y CPUs.

Tables 2 and 3 present the sweep results for FourRooms, while Tables 4, 5, and 6 present the sweep results for MultiRoom. In some cases, the value selected in the sweep is a "border value", i.e., the smallest or the largest value attempted in the sweep. While methods using border values could perform better by increasing the size of our sweep, this experimental bias is present in all evaluations, not just the baselines. This means that DIDEC's performance could also be improved in some cases.

G Selecting Masking Type

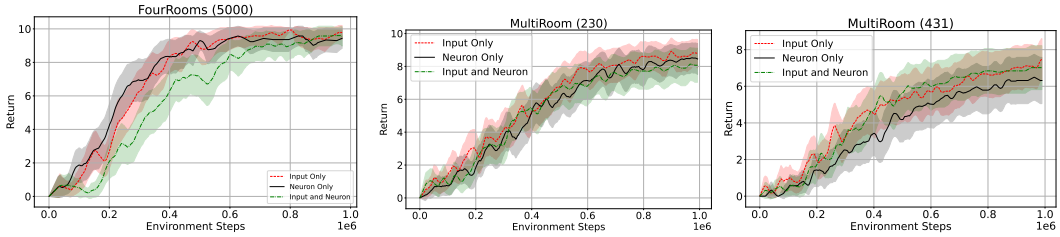


Figure 7: Evaluating masking schemes: input only, neurons only, and both. The plots present the average return and the 95% confidence intervals for 50 (FourRooms) and 57 seeds (MultiRoom).

We considered masking neurons, input features, or both as a hyperparameter. The results presented in the paper are based on the selection of the best-performing scheme for DIDEC. Figure 7 presents the performance of DIDEC when masking only the input features, only the neurons, and both. There is no noticeable difference among the different masking schema for these three environments. Note that, by choosing default values for the inputs, we select subtrees of the underlying neural tree, implicitly masking neurons. An advantage of masking neurons, which we have not explored in this work, is that it works as a compression scheme. Every neuron that is set to either active or inactive can be removed from the network, thus compressing the model.

Hyperparameter	Values Tested
Learning rate	{0.01, 0.005, 0.001, 0.0005, 0.00005}
Clipping coefficient	{0.01, 0.05, 0.1, 0.15, 0.2, 0.3}
Entropy coefficient	{0.0, 0.01, 0.02, 0.03, 0.05, 0.1, 0.2}
Training time steps	{1 million}

Table 4: Hyperparameters evaluated for DIDEC on MultiRoom.

Methods	Learning rate	Clipping coefficient	Entropy coefficient
Didec	0.01	0.2	0.01
Fine-Tune	0.01	0.3	0.02
Dec-Whole	0.001	0.3	0.02
Neural-Augmented	0.01	0.15	0.02
Vanilla	0.001	0.3	0.01

Table 5: Chosen hyperparameters per method on MultiRoom (230).

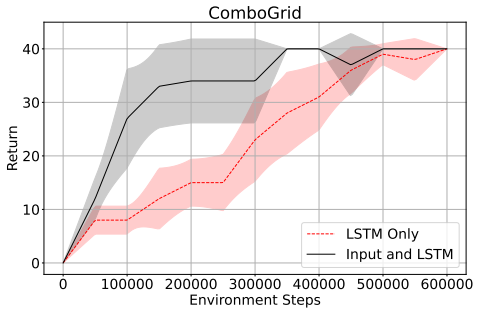


Figure 8: Comparison between learning only the initial hidden state and memory vectors of the LSTM (LSTM only), and learning the vectors and also masking the inputs.

Figure 8 shows the DIDEC results for LSTM policies on ComboGrid (dense), where one uses options learned only by learning the hidden state and memory vectors of the LSTM (LSTM Only), while the other learns the initial hidden state and memory vectors in addition to input masking (input and LSTM). These results highlight the need for both selecting the initial vectors and masking the inputs. A helpful analogy is that by learning the initial vectors, we select the initial mode of the LSTM's underlying FSM; by masking the inputs, we control which transitions we allow from that initial mode. By doing both, we are selecting a sub-automaton of the LSTM underlying FSM.

H Exploration with DIDEC's Options

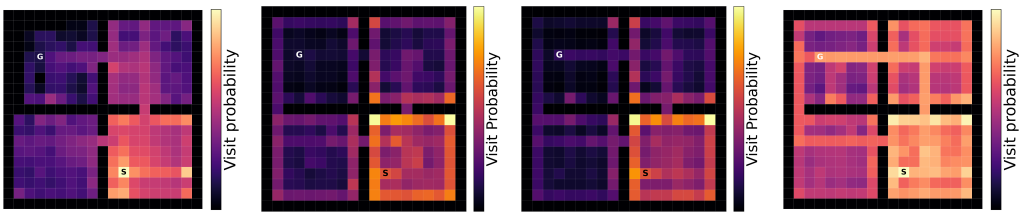


Figure 9: Probability of an agent visiting each of the cells in FourRooms in the first 500,000 training steps. From left to right Vanila, Dec-Whole, Fine-Tune, DIDEC. The data is averaged over 50 seeds.

Figure 9 presents the heatmaps denoting the probability of an agent visiting each cell in the first 500,000 training steps of the A2C agent, for Vanilla, Dec-Whole, Fine-Tune and DIDEC (from left to

Methods	Learning rate	Clipping coefficient	Entropy coefficient
Didec	0.001	0.2	0.03
Fine-Tune	0.01	0.2	0.02
Dec-Whole	0.001	0.3	0.01
Neural-Augmented	0.005	0.15	0.05
Vanilla	0.001	0.3	0.01

Table 6: Chosen hyperparameters per method on MultiRoom (431).

Hyperparameter	Values Tested
Learning rate	{0.01, 0.005, 0.001, 0.0001}
Clipping coefficient	{0.1, 0.2, 0.3}
Entropy coefficient	{0.0, 0.01, 0.02, 0.05}
Rollout Length	{128}
Training time steps	{1 million}

Table 7: Hyperparameters evaluated for DIDEC on ComboGrid - Sparse Reward.

right). DIDEC's allows for better exploration as the agent learns to navigate between rooms much more quickly than Vanilla, as evidenced by the lighter rectangle connecting the four rooms.

I Time Analysis of Option Extraction

To quantify the computational overhead introduced by option extraction and option selection, we measured wall-clock time for each system evaluated on MiniGrid SimpleCrossing and our Meta-World task under an identical compute setup. For MiniGrid, all methods run on a single CPU job with 16 physical CPU cores and 32 parallel worker processes (Python multiprocessing). For Meta-World we use a single CPU with a single process. No GPUs are used in these experiments.

For DIDEC, FineTune, and Dec-Whole we report the average and standard deviation of

1. Learning Time: time to learn all candidate options across the base problems.
2. Selection Time: time to select a subset that reduces the Levin loss.
3. Total Time: the sum of the two steps.

All results shown in Table 15. In MiniGrid, DIDEC's running time is comparable to that of Fine-Tune, but with a much improved sample efficiency (see Figure 2). DIDEC's running time of learning options is on par with the time the agent spends exploring the environment. This is loosely related to wake-sleep algorithms [Hinton et al., 1995]. In DIDEC, the agent goes through its wake cycle, where it learns to solve problems, and its sleep cycle, it organizes what it has learned into reusable options. The experiments in Meta-World show that DIDEC's neural decomposition can be used in much simpler settings, where the running time can be reduced from minutes to seconds.

J Subset Selection and Levin-Loss Computation

DIDEC can generate a large candidate set of options. Using all options in this set can slow down downstream learning because the high-level policy must learn to choose among a large number of temporally extended actions. We therefore prune the candidate set using the subset-selection criterion of Alikhasi and Lelis [2024]. This criterion is based on the Levin loss [Orseau et al., 2018], which estimates how many samples a policy would need to reproduce a set of trajectories.

Let Ω be the set of candidate options and let $\Omega' \subseteq \Omega$ be a candidate subset. For a trajectory

$$T = ((o_0, a_0), (o_1, a_1), \ldots, (o_{H-1}, a_{H-1}), o_H),$$

Hyperparameter	Values Tested
Learning rate	{0.005, 0.001, 0.0001, 0.0005}
Clipping coefficient	{0.1, 0.2, 0.3}
Entropy coefficient	{0.0, 0.01, 0.02, 0.05}
Rollout Length	{128, 256}
Training time steps	{3 million}

Table 8: Hyperparameters evaluated for DIDEC on ComboGrid - Dense Reward.

Methods	Learning rate	Clipping coefficient	Entropy coefficient
Didec	0.005	0.3	0.01
Fine-Tune	0.005	0.1	0.01
Neural-Augmented	0.005	0.1	0.01
Vanilla	0.005	0.1	0.02

Table 9: Chosen hyperparameters per method on ComboGrid - Dense Reward. The chosen rollout length for all the methods was 128.

the Levin loss of a policy π on T is

$$L(T, \pi) = \frac{H}{\prod_{t=0}^{H-1} \pi(a_t \mid o_t)}.$$

The denominator is the probability that π reproduces the trajectory, and the numerator is the number of environment interactions required by one rollout. Thus, the loss estimates the number of environment interactions required before the policy samples the trajectory.

To simulate the beginning of learning, we evaluate subsets using the uniform policy over the option-augmented action space. Let $\pi_u^{\Omega'}$ denote the uniform policy over primitive actions and options in Ω'. Adding options increases the size of the action space, which decreases the probability of each high-level choice under the uniform policy. However, useful options can cover multiple primitive actions with a single high-level decision, reducing the effective trajectory length. Levin loss balances these two effects.

When computing the loss on a trajectory T_j generated by policy π_j, we follow prior work and do not allow options extracted from π_j to be used in the computation. This prevents the selection procedure from choosing options that reproduce the policy that generated the trajectory, which is unlikely to generalize to downstream tasks. Let Ω'_{-j} denote the subset of options in Ω' whose source policy is not π_j. The score of a subset is

$$\mathrm{Score}(\Omega') = \sum_j L(T_j, \pi_u^{\Omega'_{-j}}).$$

Computing the Levin loss. The main step in subset selection is computing the Levin loss of a trajectory under a fixed option set. This requires deciding which options should be used at which points of the trajectory. Algorithm 2 gives the dynamic-programming procedure used by Dec-Options. The table M stores the minimum number of high-level decisions needed to reproduce each prefix of the trajectory. A primitive action advances the trajectory by one step, while an applicable option advances it by the duration of the option.

An option $\omega = \mathrm{repeat}(z_\omega) : \pi_\omega$ is applicable at time t if it reproduces the next z_ω actions in the trajectory, that is, if the actions selected by π_ω on $o_t, o_{t+1}, \ldots, o_{t+z_\omega-1}$ match $a_t, a_{t+1}, \ldots, a_{t+z_\omega-1}$.

Greedy subset selection. The original Dec-Options procedure uses a greedy algorithm to select a subset of options. Starting with $\Omega' = \emptyset$, the algorithm repeatedly adds the option $\omega \in \Omega \setminus \Omega'$ that most decreases $\mathrm{Score}(\Omega' \cup \{\omega\})$. The process stops when no remaining option decreases the score. Algorithm 3 summarizes the procedure.

Methods	Learning rate	Clipping coefficient	Entropy coefficient
Didec	0.001	0.1	0.01
Fine-Tune	0.001	0.2	0.02
Neural-Augmented	0.001	0.2	0.02
Vanilla	0.001	0.2	0.02

Table 10: Chosen hyperparameters per method on ComboGrid - Sparse Reward.

Hyperparameter	Values Tested
Actor Learning rate	{0.0001, 0.0003, 0.0005}
Critic Learning rate	{0.0001, 0.0003, 0.0005}
Clipping Coefficient	{0.1, 0.2}
Entropy Coefficient	{0.0, 0.01,0.02, 0.05}
Training time steps	{1 million}

Table 11: Hyperparameters evaluated for DIDEC on MiniHack (30).

Stochastic hill climbing. In our experiments, we use the same Levin-loss objective, but approximate the subset-selection problem with stochastic hill climbing. A candidate solution is a subset $\Omega' \subseteq \Omega$ with $|\Omega'| \leq s_{\max}$, where $s_{\max}$ is a hyperparameter. The search starts from a randomly selected candidate subset and repeatedly moves to the best neighbor with smaller Levin loss. If no neighbor improves the loss, the search terminates. We use random restarts and return the candidate with the smallest loss across all restarts.

The neighborhood function samples v options from $\Omega \setminus \Omega'$. If $|\Omega'| < s_{\max}$, the neighborhood includes candidates obtained by adding one sampled option to Ω'. It also includes candidates obtained by replacing one option in Ω' with one sampled option, and candidates obtained by removing one option from Ω'. To bias the search toward complementary options, the v options are sampled according to how often they can be initiated at trajectory positions not already covered by the current subset. A trajectory position is uncovered if, in Algorithm 2, the dynamic program reaches the next state through a primitive action rather than through an option. This neighborhood therefore favors options that cover parts of the training trajectories not yet covered by the current subset.

K Finite-Class Guarantee

Although the number of discrete masked options is exponential, this does not mean that empirical imitation loss is statistically uninformative. The theorem below gives a standard finite-class guarantee for the discrete mask class: after searching over masks, a mask with low empirical imitation loss is unlikely to be a lucky fit to the sampled sub-trajectories. This result should be interpreted as a guarantee for the mask class itself, not as an optimization guarantee for DIDEC's straight-through approximation. The bound is meaningful for any discrete mask returned by the learning procedure. However, it does not imply that gradient descent will find a mask that minimizes the empirical loss.

Fix a policy $\pi_\ell \in \Pi_{\text{train}}$ with hidden ReLU units H_ℓ, where $|H_\ell| = d_\ell$, and binary observations $X \in \{0,1\}^{n_1}$. Let $\mathcal{M}_X = \{\text{read}, 0, 1\}^{[n_1]}$ and $\mathcal{M}_{HX}(\pi_\ell) = \mathcal{M}_H(\pi_\ell) \times \mathcal{M}_X$ be the input-mask space and the joint hidden-input mask space. Thus $|\mathcal{M}_{HX}(\pi_\ell)| = 3^{d_\ell + n_1}$. For a discrete mask configuration $M \in \mathcal{M}_{HX}(\pi_\ell)$, let π_ℓ^M be the corresponding masked policy. Let $\mathcal{D}_b$ be a distribution over length-b sub-trajectories, and let $\ell(\pi_\ell^M, \tau) \in [0,1]$ be a bounded imitation loss on τ. Define

$$L_{\mathcal{D}_b}(M) = \mathbb{E}_{\tau \sim \mathcal{D}_b}\left[\ell(\pi_\ell^M, \tau)\right] \quad \text{and} \quad \widehat{L}_n(M) = \frac{1}{n}\sum_{r=1}^{n} \ell(\pi_\ell^M, \tau_r).$$

Proposition 1. *If $\tau_1, \ldots, \tau_n$ are drawn independently from $\mathcal{D}_b$, then with probability at least $1-\delta$, every mask $M \in \mathcal{M}_{HX}(\pi_\ell)$ satisfies*

$$L_{\mathcal{D}_b}(M) \leq \widehat{L}_n(M) + \sqrt{\frac{(d_\ell + n_1)\log 3 + \log(2/\delta)}{2n}}.$$

Methods	Actor learning rate	Critic learning rate	Entropy coefficient
Didec	0.0001	0.0005	0.05
Fine-Tune	0.0005	0.0003	0.02
Dec-Whole	0.0001	0.0005	0.02
Neural-Augmented	0.0003	0.0005	0.02
Vanilla	0.0005	0.0005	0.02

Table 12: Chosen hyperparameters per method on MiniHack (30).

Hyperparameter	Values Tested
Learning rate	{ 0.0008, 0.0003, 0.0001, 0.00005}
Entropy coefficient	{0.0, 0.001, 0.005}
Training time steps	{200000}

Table 13: Hyperparameters evaluated for baselines on Meta-World.

Proof. Fix $M \in \mathcal{M}_{HX}(\pi_\ell)$ and define $Y_r = \ell(\pi_\ell^M, \tau_r)$. The random variables $Y_1, \ldots, Y_n$ are independent, take values in $[0, 1]$, and have mean $L_{\mathcal{D}_b}(M)$. Hoeffding's inequality gives

$$\Pr\left(\left|\widehat{L}_n(M) - L_{\mathcal{D}_b}(M)\right| > \epsilon\right) \leq 2\exp(-2n\epsilon^2).$$

Applying a union bound over the finite class $\mathcal{M}_{HX}(\pi_\ell)$ gives

$$\Pr\left(\exists M \in \mathcal{M}_{HX}(\pi_\ell) : \left|\widehat{L}_n(M) - L_{\mathcal{D}_b}(M)\right| > \epsilon\right) \leq 2\,|\mathcal{M}_{HX}(\pi_\ell)|\exp(-2n\epsilon^2).$$

Since $|\mathcal{M}_{HX}(\pi_\ell)| = 3^{d_\ell + n_1}$, choosing

$$\epsilon = \sqrt{\frac{(d_\ell + n_1)\log 3 + \log(2/\delta)}{2n}}$$

makes the right-hand side equal to δ. With probability at least $1 - \delta$, the bound therefore holds for all masks simultaneously, which implies the one-sided inequality in the theorem. □

Thus, if DIDEC finds a mask with low empirical imitation loss, its population imitation loss is controlled by a term that scales with $(d_\ell + n_1)\log 3$, not with $3^{d_\ell + n_1}$. The theorem is stated for binary observations and discrete input-default masks. For continuous inputs, the same argument applies only to the discrete read/default decisions after the default values are fixed; bounding the effect of learning continuous default values would require additional assumptions.

L Problem Domains

MiniGrid. We consider eight Simple-Crossing configurations as training problems and FourRooms as the test problem. We also use three MultiRoom environments for training and a fourth for testing. MultiRoom is more challenging than FourRooms because the agent must find a key and unlock a door, and because room shapes vary, making generalization harder. We adopt MiniGrid's original action space (turn left, right, move) and an egocentric 9×9 observation window.

MiniHack. We consider the corridors challenge of MiniHack, where the agent learns to navigate in long procedurally generated rooms and corridors of the game NetHack [Küttler et al., 2020]. This environment offers dynamics and challenges different from MiniGrid, such as the agent starving. We consider four environments as training, and a fifth environment with longer corridors and more rooms as the test problem. As in MiniGrid, we also use an egocentric view of size 9×9 around the agent.

ComboGrid. In ComboGrid, the agent must learn action sequences whose effects occur only after completing an "action combo". We consider three primitive actions (0, 1, 2) and three combos: 0,0,1 (turn right); 1,1,0 (turn left); and 2,2,1 (move forward). Because observations change only after a combo, the agent must remember sequences. Training uses four 5×5 grids surrounded by walls, leaving a 3×3 area where the agent collects a marker. For testing, we remove the walls to yield a full 5×5 grid with four markers. We study two reward settings: dense, where the agent receives a positive reward per marker, and sparse, which rewards only after collecting all markers.

Methods	Learning rate	Entropy coefficient
Didec	0.0003	0.001
Fine-Tune	0.0003	0.0
Dec-Whole	0.00005	0.001
Neural-Augmented	0.00005	0.001
Vanilla	0.0001	0.005

Table 14: Chosen hyperparameters per method on Meta-World.

MiniGrid			
Method	**Learning**	**Selection**	**Total**
DIDEC	1476.3 ± 760.4	1359.4 ± 631.1	2835.7 ± 1390.3
Fine-Tune	871.0 ± 464.9	1387.1 ± 668.8	2258.1 ± 1131.0
Dec-Whole	0.0 ± 0.0	53.4 ± 17.8	53.5 ± 17.8
Meta-World			
Method	**Learning**	**Selection**	**Total**
DIDEC	10.21 ± 0.05	-	10.21 ± 0.05
Fine-Tune	5.28 ± 0.02	-	5.28 ± 0.02
Dec-Whole	0.0 ± 0.0	-	0.0 ± 0.0

Table 15: Average running time in seconds and standard deviation for option learning and selection across different approaches. Dec-Whole does not perform any training, so the learning time is zero. The Meta-World experiments do not include a selection step because we use one option per base policy. We use 50 seeds for MiniGrid and 40 for Meta-World.

Meta-World. In Meta-World, the agent controls a robotic arm to perform various tasks. As observations, the agent receives the coordinates of different objects in 3D space; as actions, it can control the direction of the robotic arm's gripper, as well as opening and closing it. Both observations and actions are continuous. In our experiments, we consider two training problems where the agent moves a plate into a hockey net. The training problems differ in the position of the arm, plate, and net. The test problems are more difficult as the distance between the plate and the net changes, one by 0.85 times the distance in the training problems (Closer Goal) and the other by 1.35 times the distance (Farther Goal). Since selecting a subset of options while minimizing the Levin loss is inherently discrete, we use a simplified version of DIDEC with Meta-World to handle its continuous spaces. When training masks, instead of considering all sub-trajectories of each $\mathcal{T}_j$ of $\mathcal{T}_{\text{train}}$, we consider only $\mathcal{T}_j$, thus resulting in one option per trajectory. Learned options are made available to the agent through a mixture-of-experts scheme [Jacobs et al., 1991] (Appendix M), thus bypassing the selection step. The experiments in Meta-World serve two purposes: to show that our mask-based decomposition can also be used in continuous spaces and in a simpler setting that bypasses the subset selection step.

Figure 10 shows the ten SimpleCrossing environments used in training, and the FourRooms environments used in testing. Figure 11 shows the training and test problems for MultiRoom. Interestingly, DIDEC outperforms the baselines by a large margin on problem 230 (Figure 2), whose middle room has a structure different than the training problems—the room is narrower than the rooms in problems 1, 3, and 17. This discrepancy can affect generalization because the agent has not previously observed this type of room. This result demonstrates the generalization capabilities of the learned options. For both FourRooms and MultiRoom, we use MiniGrid's standard reward function [Chevalier-Boisvert et al., 2023].

Figure 12 shows the four ComboGrid training problems and the test problem, where the agent needs to collect all four markers to complete the task. In the dense version of the environment, each marker gives a reward of ten; any other step gives zero reward. In the sparse version, the agent receives one after collecting all markers, and zero otherwise.

Algorithm 2 COMPUTE-LEVIN-LOSS

Require: Trajectory $T = ((o_0, a_0), \ldots, (o_{H-1}, a_{H-1}), o_H)$, option set Ω, primitive action set A
Ensure: $L(T, \pi_u^\Omega)$

$p_{u,\Omega} \leftarrow 1/(|A| + |\Omega|)$
$M[j] \leftarrow j$ for $j = 0, 1, \ldots, H$
for $j = 0, 1, \ldots, H-1$ **do**
  $M[j+1] \leftarrow \min(M[j+1], M[j]+1)$
  for each option $\omega \in \Omega$ **do**
    Let z_ω be the duration of ω
    if $j + z_\omega \leq H$ and ω is applicable at o_j **then**
      $M[j+z_\omega] \leftarrow \min(M[j+z_\omega], M[j]+1)$
return $H \cdot (p_{u,\Omega})^{-M[H]}$

Algorithm 3 GREEDY-SUBSET-SELECTION

Require: Candidate options Ω, training trajectories $\mathcal{T}_{\text{train}}$, primitive action set A
Ensure: Selected subset Ω'

$\Omega' \leftarrow \emptyset$
$l \leftarrow \text{Score}(\Omega')$
while true **do**
  $\omega^\star \leftarrow$ None, $l^\star \leftarrow l$
  for each option $\omega \in \Omega \setminus \Omega'$ **do**
    $c \leftarrow \Omega' \cup \{\omega\}$
    $l_c \leftarrow \text{Score}(c)$
    if $l_c < l^\star$ **then**
      $\omega^\star \leftarrow \omega$, $l^\star \leftarrow l_c$
  if $\omega^\star =$ None **then**
    break
  $\Omega' \leftarrow \Omega' \cup \{\omega^\star\}$
  $l \leftarrow l^\star$
return Ω'

Figure 13 shows the training and test problems of the Corridors environment we used from MiniHack. In this environment, the agent receives a reward of -1 for every step, except for the transition to the stairs, when the episode terminates.

M Mixture of Experts for Continuous Action Spaces

To show a broader applicability of DIDEC's decomposition, we also apply it to a Meta-World task [Yu et al., 2021], which has continuous actions and state spaces. Instead of performing subset selection, for domains with continuous actions, we use a mixture-of-experts approach [Jacobs et al., 1991], combining the primitive actions with that of one or more options. The policy outputs observation-dependent means and log standard deviations, and actions are sampled with the reparameterization trick:

$$a = \mu(o) + \sigma(o) \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

where $\mu(o)$ and $\sigma(o)$ are the outputs of the policy for o.

Given a set of primitive and options of size K, the agent learns to mix them by learning K parameters θ.

$$a_{\text{mix}} = \sum_{i=1}^{K} \theta_i a_i, \qquad \theta_i \geq 0, \quad \sum_{i=1}^{K} \theta_i = 1\,.$$

Here, a_{mix} is the action the agent takes, and the mixture weights θ are produced with a Softmax network and learned end-to-end while the agent learns to solve a given POMDP.

Unlike options [Sutton et al., 1999], this mixture-of-experts approach does not produce temporally extended actions: each subprogram provides an action at every timestep.

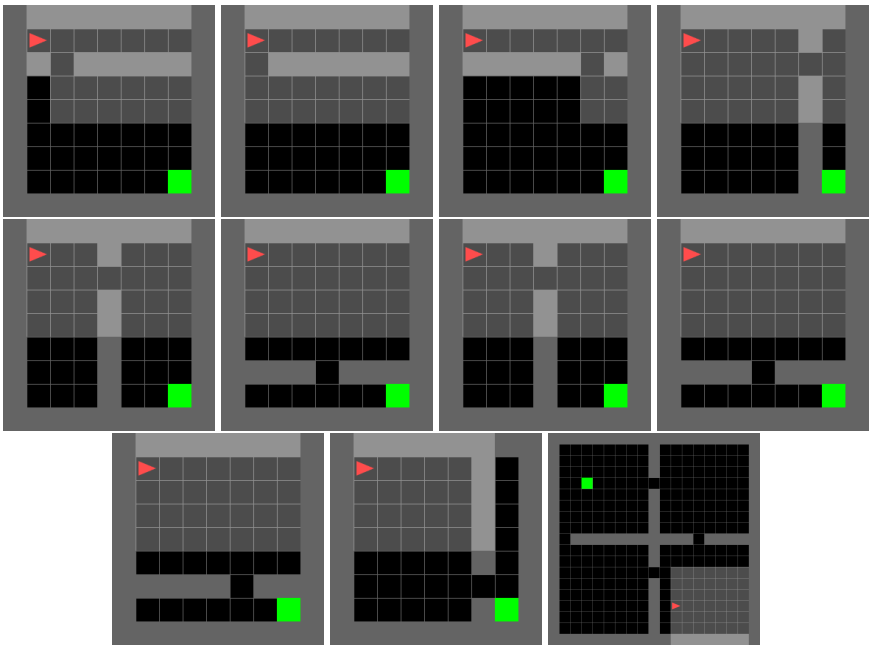
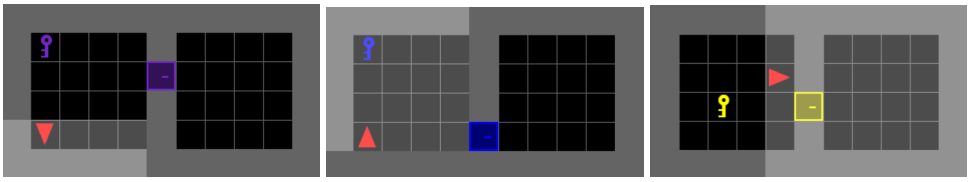
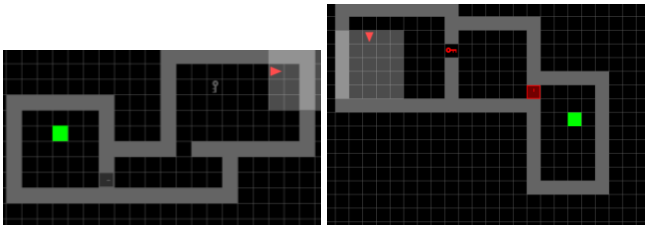


Figure 10: SimpleCrossing problems used as training for the FourRooms experiments. The seeds used to generate the problems, from left to right and top to bottom: 1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, 10000. The bottom-right image shows the FourRooms environment used (5000).



(a) Training problems used in MultiRoom. The seeds used to generate the problems, from left to right: 1, 3, 17.



(b) Test problems used in MultiRoom. The seeds used to generate the problems, from left to right: 230 and 431.

Figure 11: Training and test problems used in our MultiRoom experiments.

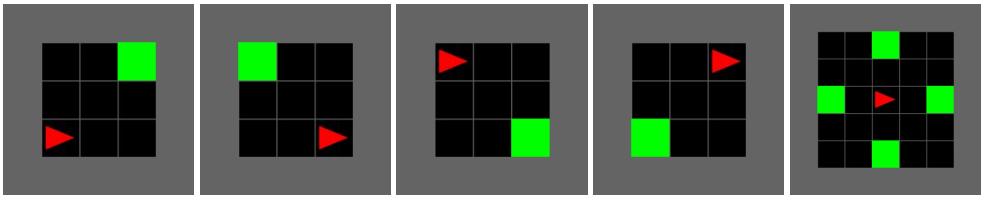


Figure 12: The four ComboGrid training environments, from left to right and top to bottom, and the test environment.

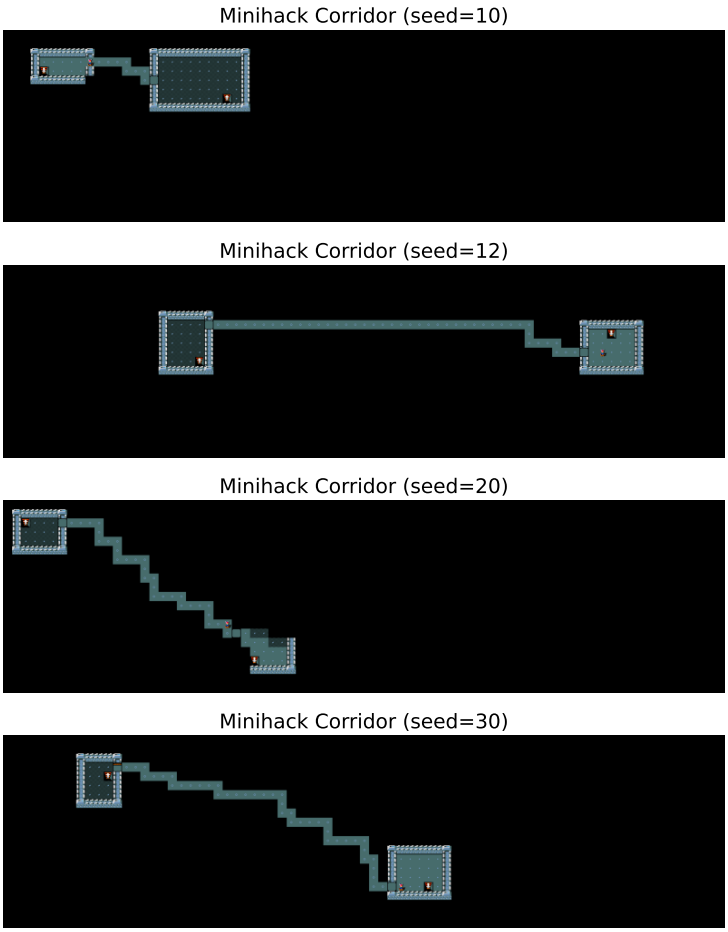


Figure 13: The three MiniHack Corridor training environments (top), and the test environment (bottom), generated with seeds 10, 12, 20, and 30, respectively.

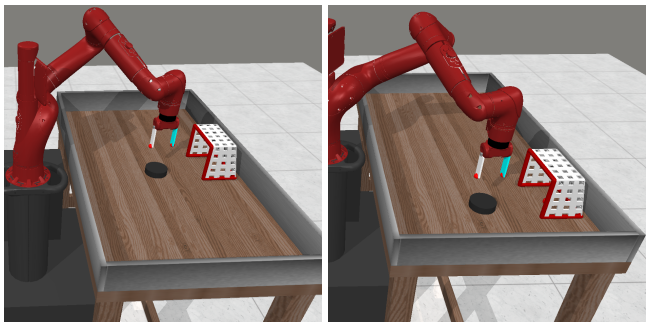


Figure 14: The two Meta-World training environments. In addition to the location of the net, the plate, and the arm, the distance between the plate and the net was changed in the test problems. The plate is closer to the goal in one test problem and farther in the other.

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